CALIBER 2026 | BUSINESS PROPOSAL | PRELIMINARY STAGE

Asset Knowledge Platform

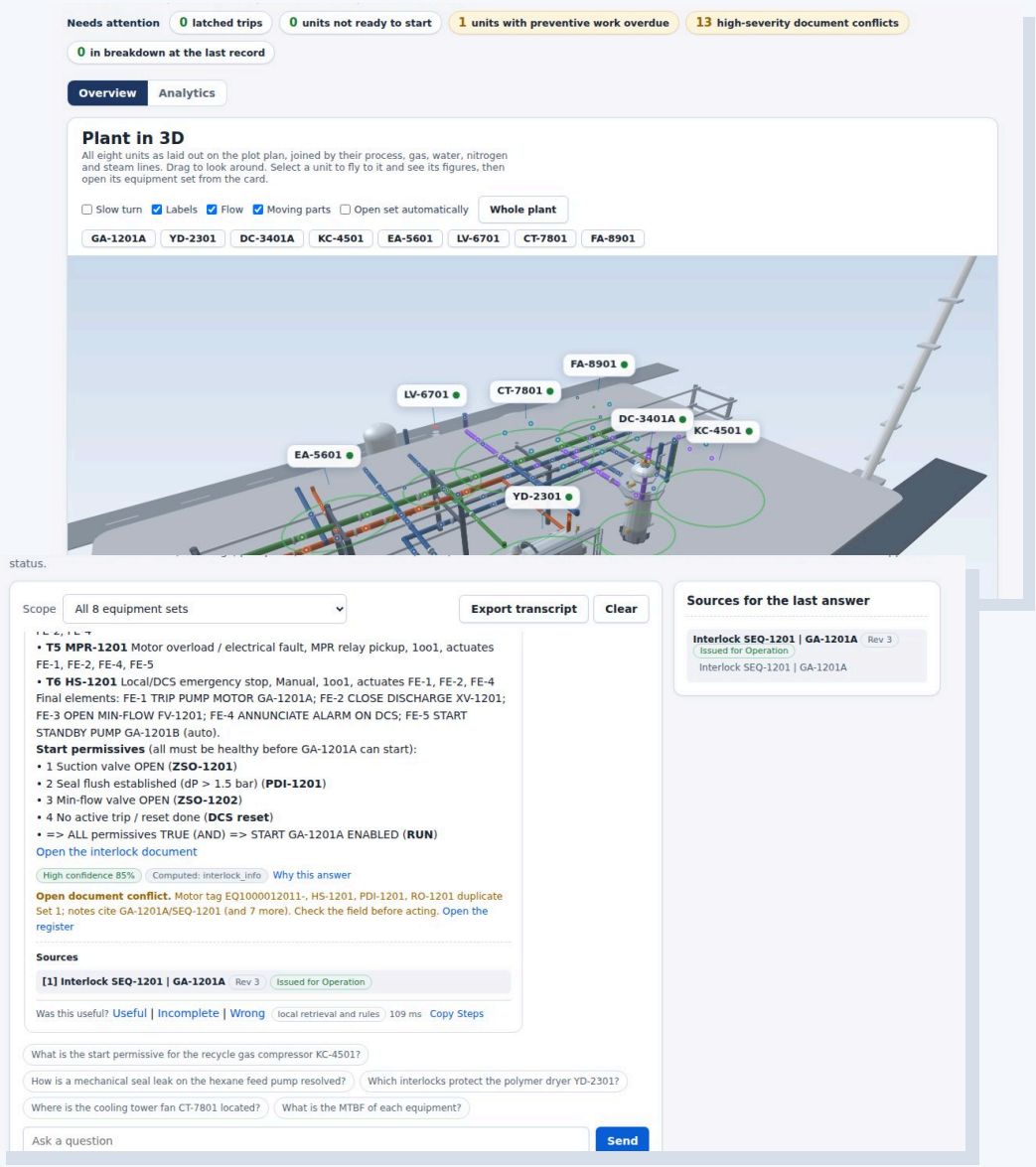
One trusted, traceable knowledge hub for the plant, with an AI assistant that shows its sources, and knowledge that is put to work in daily decisions.

Chosen use case: Case 1, Manufacturing Knowledge Hub

98	211	20	128/128
documents joined	work orders read	computed AI tools	test questions passed

Team: [Team name]

University: [University] **Supervisor:** [Supervisor name]



Working prototype with the case data loaded, a 3-minute demo video and this deck

Engineers cannot find one verified, current answer fast, so routine work turns into execution risk

Background and problem statement: the case describes it, and the baseline data confirms it

The problem in the case data

98
source documents in 8 formats, none linked to each other

48
conflicts and gaps between drawings, sheets and logic

211
work orders in 1.52 years, kept as separate records

31
breakdowns, 434 h of downtime

IDR 538 M
maintenance cost, 82% of it corrective

21%
of faults found early, before they became a corrective job

How scattered knowledge becomes execution risk

ROOT CAUSES

Scattered sources
SOPs, datasheets, P&IDs, EDMS, AIMS and logs sit in different systems

Missing tacit knowledge
Why a trip happens or a workaround works stays with senior people

Conflicting documents
A correct-looking answer can be the wrong revision

Disconnected data
A work order, a trip and a drawing for one tag are never read together

WHAT HAPPENS IN THE PLANT

Longer troubleshooting
time lost searching and asking

Repeated engineering rework
same fault, same investigation

Higher safety risk
wrong step, valve position or set point

Delayed start-up
Process, Maintenance and HSE read different truths

Real example from the case drawings. Valve FV-2302 is labelled both "trips OPEN" and "trips CLOSED", and one sheet is titled KA-4501 while its body says KC-4501. An engineer who trusts the wrong one acts on the wrong valve position.

Who is affected

Operators
need the exact step and set point mid-shift

Maintenance technicians
need past failures and the right part first time

Reliability engineers
need repeat failures and root causes in one place

Process engineers
need trusted limits and design basis

HSE officers
need hazards and permits tied to the asset

Supervisors, managers
need one answer, not three reports

Problem statement: scattered, partly conflicting plant knowledge makes engineers search, ask around or guess, which is how improper execution happens.

Source: CALIBER 2026 case book (Case 1); baseline data analysed in the prototype (8 equipment sets, LLDPE unit, Jun 2024 to Dec 2025)

Asset Knowledge Platform: connect the sources, answer with proof, and put the knowledge to work

Solution: objective, workflow and key technologies. It answers the three key questions of the case (Q1, Q2, Q3)

Main problem

48 **conflicts across 98 plant documents**

No single place engineers can trust, so they search, ask or guess.

Objective

1. **Connect** scattered knowledge into one tag-keyed, governed base
2. **Answer** with source, version, status and confidence
3. **Act** use failure memory, prediction and hazards in daily work

Implications

Engineers	▼ Search time	▼ Wrong steps
Plant	▲ Reliability	▼ Late start-up

CONNECT Industrial Data Ops — Q1

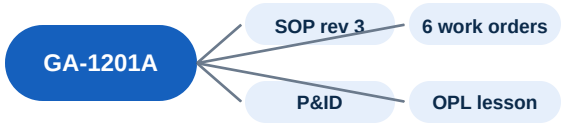
Build one structured, connected knowledge base from every plant source

1. INGEST EVERY SOURCE

SOP · Datasheet · P&ID · Interlock C&E · Maint. log · Incident / RCA · OPL

AI reads each file and pulls out tags, steps and limits

2. JOIN THEM ON THE EQUIPMENT TAG



3. CHECK, APPROVE, GOVERN

Draft ▶ In review ▶ Approved

48 conflicts found on day one, each with an owner. Secrets masked, access by role.

ASK Trusted answers — Q2

Help engineers find trusted answers fast and avoid improper execution

1. ASK IN PLAIN LANGUAGE

"What must be healthy before the hexane feed pump GA-1201A can start?"

Role and equipment in view set the scope

2. FIND AND COMPUTE

Keyword + meaning search · **20 computed tools** · **Approved sources only**

3. ANSWER WITH PROOF

4 permissives: suction valve open, seal flush dP above 1.5 bar, min-flow valve open, no active trip [1]

SEQ-1201 Rev 3 · Issued for Operation · Confidence 85%

No evidence, no answer. Interlock bypass refused.

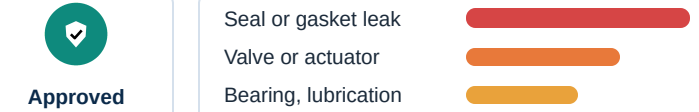
ACT Active knowledge — Q3

Give reliability, troubleshooting and improvement active decision support

1. LEARN FROM PAST FAILURES

WO-240088	Crosshead vibration
RCA	Pin clearance too wide
Lesson	OPL: check clearance at 6 months

2. PREDICT AND PREPARE



Risk ranked; start-up readiness GO / NO-GO

3. PLUG INTO PLANT SYSTEMS

EDMS · AIMS · Digital twin · Historian

Read-only links. Site planner, replay and hazard simulator use the same tag-keyed base.

KEY TECHNOLOGIES

- **Hybrid retrieval** — keyword + meaning
- **LLM + tool calls** — any provider
- **Tag-keyed facts** — typed links
- **Reliability maths** — Weibull, Poisson
- **Serverless store** — Netlify, Postgres
- **Governance** — roles, audit

Source: Prototype built on the case baseline data; counts from the running application

Base case: about IDR 1.5 B of value a year, paid back in roughly 21 months, with downside shown

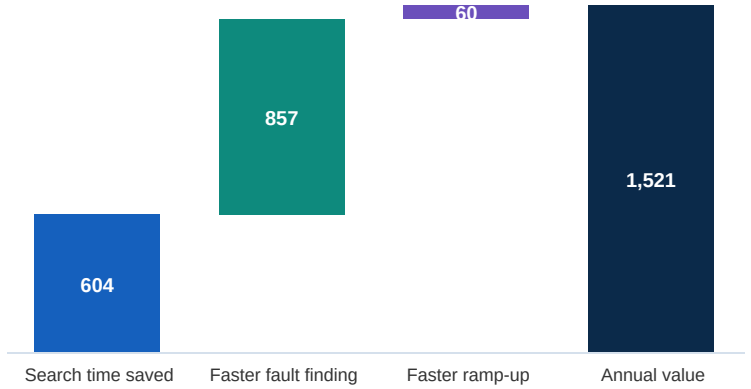
Business impact: measurable benefits, KPIs and success metrics (assumptions marked; replace with Company figures during the pilot)

KPIs and success metrics

KPI	TODAY	TARGET
Time to a verified answer	~15 min (assumed)	3 min
Answers with source, version, status	manual	100%
Breakdown repair time (MTTR)	14 h	12.6 h
Repeat failures within 120 days	8 of 55	-30%
Faults found before failure	21%	35%
Open document conflicts	48	90% closed
Lessons captured (OPL)	56	+12 a quarter

Today values marked in the data come from the 211 work orders

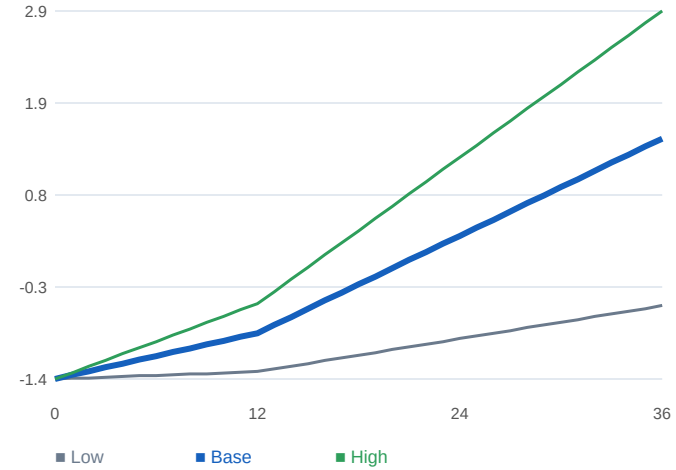
Annual value by lever, base case (IDR M)



Low case	Base case	High case
IDR 0.8 B	IDR 1.5 B	IDR 2.1 B
a year	a year	a year
no payback in 3 y	payback 21 mo	payback 16 mo

Levers: 45 users saving 20 min a day, repair time cut 10%, and one month faster ramp-up for 4 hires a year. Low and high cases halve or raise each lever.

Cumulative net benefit (IDR B)



IDR 1.4 B	IDR 370 M
one-off cost	run cost a year
21 months	58%
payback, base	3-year ROI, net IDR 1.4 B

90-day pilot gates: 90%+ answers with an approved source | 3 min median time to an answer | 80%+ users rate answers useful | 0 answers without a source. **Not counted:** avoided incidents, spare-part savings, knowledge saved from retirements.

Source: Case baseline data (work orders, MTTR, breakdowns); assumptions in appendix A8. IDR = Indonesian rupiah

Feasible now: the hub is advisory and read-only toward plant systems, and rolls out behind four decision gates

Feasibility across six readiness areas, and an implementation roadmap from the November 2026 finals to plant-wide use

Technical

Ready

- Prototype runs end to end on case data
- SQL schema and API spec written
- Model is swappable by contract

Operational

Ready

- Advisory only, no write to DCS
- Works offline with local search
- Fits OPL and work-order routines

Legal

Decision needed

- Enterprise AI terms, no training on our data
- Data held in region (PDP Law)
- Drawing and vendor IP respected

Cybersecurity

Planned

- Hub outside the OT zone (IEC 62443)
- Roles, audit trail, secret masking
- Prompt-injection guards, rate limits

Data governance

Ready

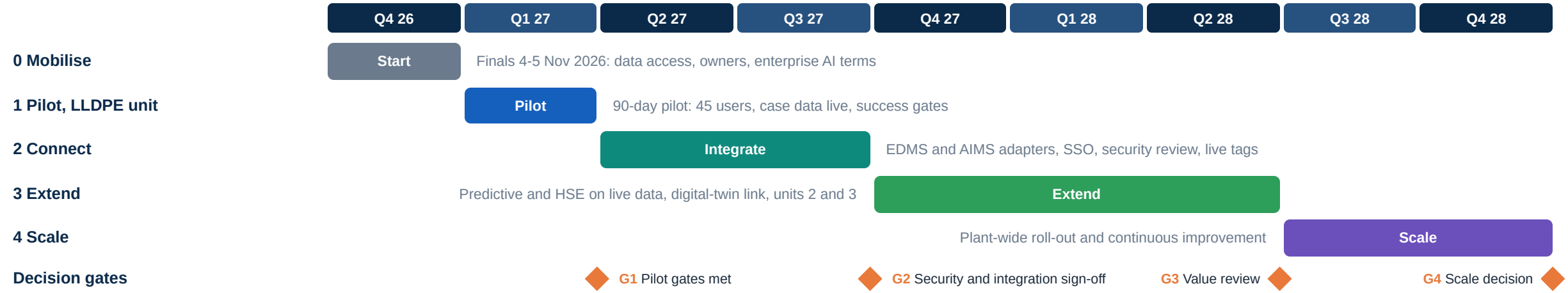
- Owner, version, status on every file
- 48 conflicts held in a register
- Approval before anything is trusted

Organisation

Planned

- A champion in each function
- Two-hour role-based training
- Monthly KPI review

Implementation roadmap



Main risks and mitigations

- **Wrong answer:** sources, verification, refusal rules
- **Poor data:** conflict register, owners
- **Low adoption:** champions, daily-use tools
- **Model cost or outage:** swappable model, offline fallback

Source: Prototype and architecture notes; roadmap dates are proposals for discussion with the Company

Engineers get one verified answer in minutes, and the plant keeps what its people know

Conclusion: the proposal, how it answers the three key questions of the case, and what we ask for next

CONNECT One governed, tag-keyed base

- 98 documents and 211 work orders joined
- 48 conflicts surfaced and owned
- Ingest, check, approve for every file

98 to 1 documents into one governed base

ASK Answers that show their proof

- Source, revision, status, confidence
- 128 of 128 test questions passed
- Bypass requests refused

128/128 questions answered with proof

ACT Knowledge in daily decisions

- Failure memory with RCA and lessons
- Risk ranking, readiness, hazards, site plan
- Read-only EDMS, AIMS, twin links

20 tools computed from live records

How the proposal answers the three key questions of the case

Q1	**Build a structured Industrial Data Ops foundation**	Tag-keyed data model over 7 source types; ingest, check and approve with owners; conflicts found on day one	**Evidence:** 98 docs, 1,109 passages, 48 conflicts logged
Q2	**Help engineers find trusted answers faster, reduce improper execution**	Hybrid search; cited answers with revision, status, confidence; verification of quotes and numbers; refusal rules	**Evidence:** 128/128 questions; 3 min target vs ~15 min
Q3	**Integrate with operational systems for reliability and improvement**	Read-only EDMS, AIMS, historian, twin links; failure memory, predictive, readiness, hazard and site planning	**Evidence:** 64-forecast backtest; 20 computed tools

The ask: a 90-day pilot

Weeks 1-2
Agree data access, owners and AI terms

Weeks 3-6
Load the LLDPE unit, train 45 users

Weeks 7-12
Run to the success gates, then decide

Source: Prototype test results (appendix A9) and business case (slide 4, appendix A8)

Team profile

Team name, university, supervisor and members (replace the bracketed fields before submitting)

Team name: [Team name] **University:** [University] **Supervisor:** [Supervisor name]

No.	Name	Major	Semester	Area of expertise	Contribution to the proposal
1	[Name]	[Computer Science]	[6]	[AI and software]	[Formulated the idea, built the prototype and the assistant]
2	[Name]	[Chemical Engineering]	[7]	[Process and reliability]	[Background and problem, process content, hazard models]
3	[Name]	[Industrial Engineering]	[7]	[Maintenance and business case]	[Business impact, KPIs, roadmap]

The booklet allows two to three members from the same university. Delete unused rows.

A1 System architecture: five layers on one tag-keyed knowledge core

How data flows from plant sources to the people who use it, with governance across every layer

1 Sources	2 Industrial Data Ops	3 Knowledge core	4 AI services	5 Experiences
SOPs, OPLs, tacit notes	AI document ingestion	Documents and chunks	Question router	AI assistant
Datasheets	Structure and tag mapping	Search index (BM25 + vectors)	Retrieval and re-ranking	Equipment 360, 2D and 3D
P&ID, GA, plot plan	Quality gates, conflicts	Facts: tags, set points, parts	20 computed tools	P&ID, interlock, OPL simulators
Interlock logic (C&E)	Secret masking	Relations between records	LLM drafting, tool calls	Maintenance and failure memory
Maintenance logs, RCA	Owner, version, status	Failure memory, lessons	Verification and confidence	Site planner and replay
Incident reports	Approval workflow	Answer, audit and feedback log	Reliability and hazard models	Hazard and readiness
Live data (DCS, historian)				

Cross-cutting governance: roles and access | approval before trust | audit trail | prompt-injection guards | retention and region | read-only toward plant systems

Prototype today: Single-file web app, Netlify functions and Blobs, free Gemini model, local fallback search

Target production: Postgres with pgvector, queue for ingestion, SSO, EDMS and AIMS adapters, Company-approved model

Source: Asset Knowledge Platform architecture notes (docs/ARCHITECTURE.md in the prototype repository)

A2 How the AI assistant answers: route, retrieve, compute, draft, verify

Every step is logged; the language model only words what the evidence already says

Browser (the app)

1 Question, role, equipment in view
Context comes from the page the user is on

2 Router
Picks a computed tool (KPI, forecast, hazard) or search

3 Retrieve
Keyword and meaning search; approved sources, role-limited

4 Compute
Tools calculate figures from work orders and records

7 Verify and show
Numbers and quotes checked against source text; confidence scored; sources, status and revision listed

Serverless function

5 Proxy
Rate limits; the key stays on the server

Prompt
Question, numbered passages, tool results, rules

Stream back
Tokens reach the page as they arrive

Fallback
If the model fails, local retrieval answers

Language model

6 Draft
Writes only from the passages given, with [n] markers

Tool calls
May ask for a tool by name; the page runs it

Any provider
Gemini free tier today; contract allows others

Data and logs

Knowledge store
Documents, versions, approvals

Answer log
Question, sources, confidence

Feedback queue
Useful, incomplete, wrong

Audit trail
Who, what, when

Evaluation set
128 questions run on every build; fails the build on a low score

Guardrails: no evidence, no answer | safety bypass requests refused with the reason | conflicting or draft sources flagged on the answer | retrieved text treated as data, never as instructions

Source: Prototype assistant pipeline (docs/AI_PROVIDER_CONTRACT.md, docs/PROMPT_TEMPLATE.md)

A3 Data sources: every source has a purpose, an owner and a business reason

Baseline data from the case, plus sources we propose to add; a few trusted sources connected well beat many connected badly

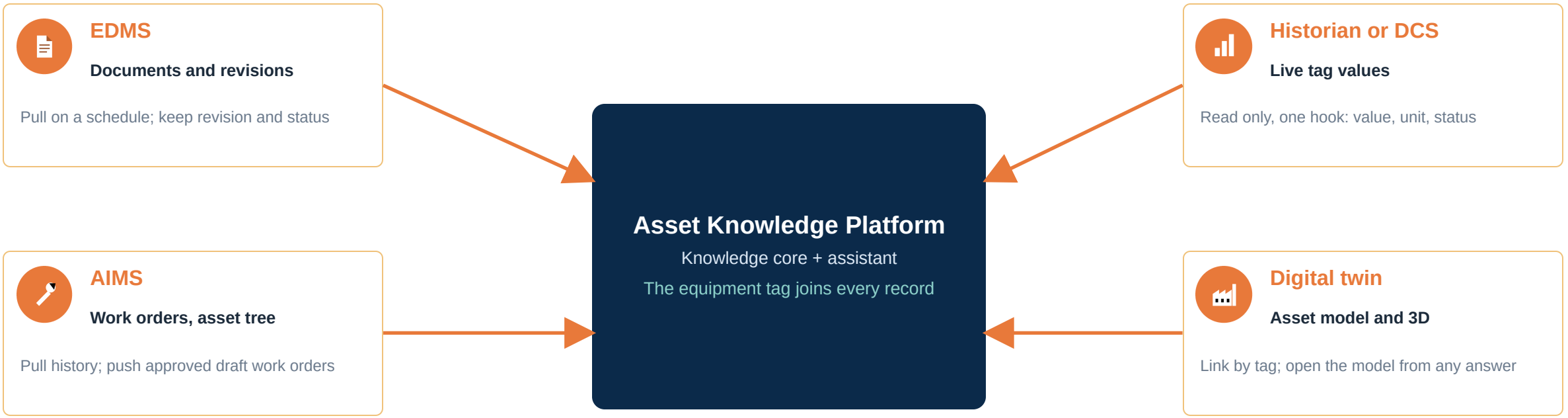
Source	What it gives	Business justification	Status
SOPs and OPLs (56)	Steps, safety notes, tacit lessons	Fewer wrong steps; keeps SME experience	Loaded
Datasheets (8)	Design limits, materials, ratings	One place for trusted specifications	Loaded
P&IDs, GA drawings, plot plans (24)	Equipment, tags, layout, hazards	Ties every tag to its drawing and location	Loaded
Interlock logic (8 sets, 84 instruments)	Causes, effects, permissives	Answers what trips and why, with the exact set point	Loaded
Maintenance history (211 WOs)	Failures, root causes, cost, downtime	Failure memory, KPIs, prediction, site planning	Loaded
Incident and RCA reports	Lessons learned	Prevents repeat events	Add in pilot
EDMS	Controlled document revisions	Single source of truth for versions	Adapter ready
AIMS	Work orders and asset hierarchy	Live history; draft work orders go back for approval	Adapter ready
Historian or DCS values	Live tag values	Context for troubleshooting and prediction	Hook in place
Digital twin	Asset model and 3D	Navigate by equipment and place	Planned

Every added source needs an owner, a refresh rule and a reason. The case asks for a well-justified data architecture rather than more data, so each row above states what decision it improves.

Source: Case book data table; prototype counts

A4 Integration: EDMS, AIMS, digital twin and plant data feed one core, read-only first

Same contract for every source: a new system is one adapter, with no change to the knowledge core



Single sign-on
Roles come from the company directory

Read first, write later
Writes are drafts that a person approves

One contract per source
New source = one adapter

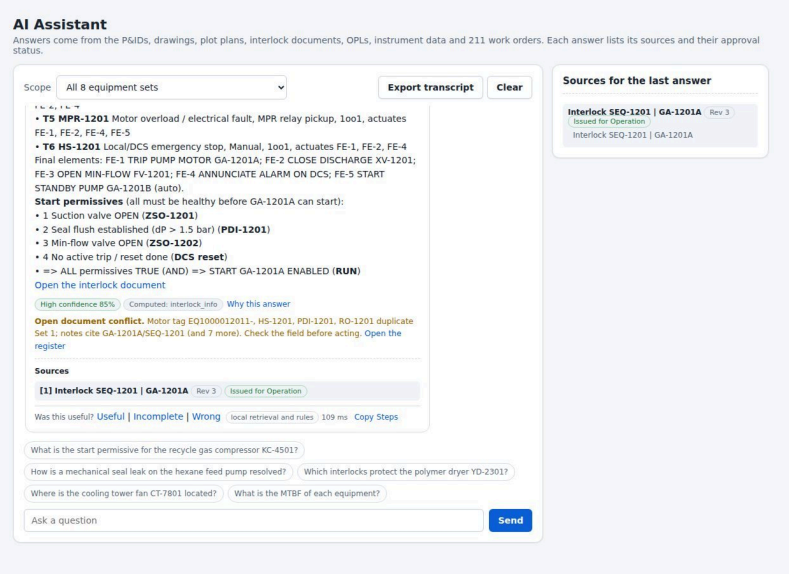
Data stays in the tenant
Region and retention set by the Company

Source: Architecture notes; adapters are specified, not yet connected to Company systems

A5 Ask, see the plant, open the equipment

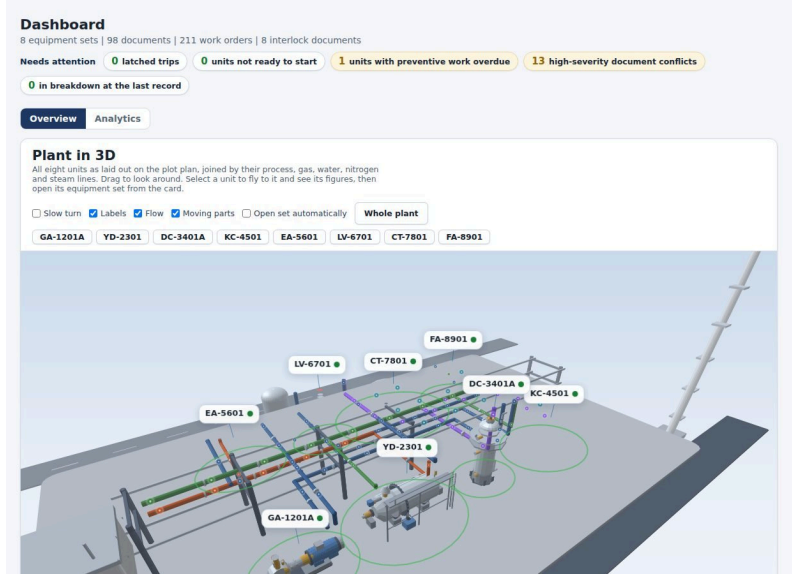
Prototype tour 1 of 3: assistant, plant overview and equipment in 3D

1 Ask and verify



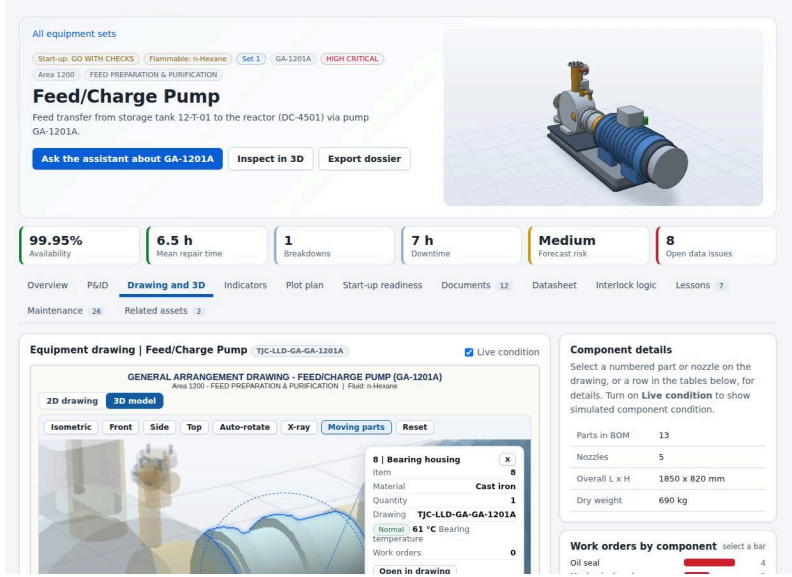
Plain-language question about start permissives. The answer lists each permissive with its tag, then the source (SEQ-1201 Rev 3, Issued for Operation), a confidence level and a warning that an open document conflict touches this equipment.

2 Plant in 3D



All eight units on the plot plan with live state. Attention strip on top: latched trips, units not ready to start, overdue preventive work, high-severity document conflicts. Select a unit to fly to it and open its set.

3 Equipment in 3D



The equipment drawing as a 3D model. Select a part to see what the drawing says about it, its rating, and the work orders that mention it. Part names match the datasheet and the P&ID.

Why it matters: Answers key questions Q2 and Q3: trusted answers found fast, and knowledge shown in the context of the asset and its operating state.

Source: Screens from the running prototype with the case data loaded

A6 Maintenance: replay, site plan and failure outlook

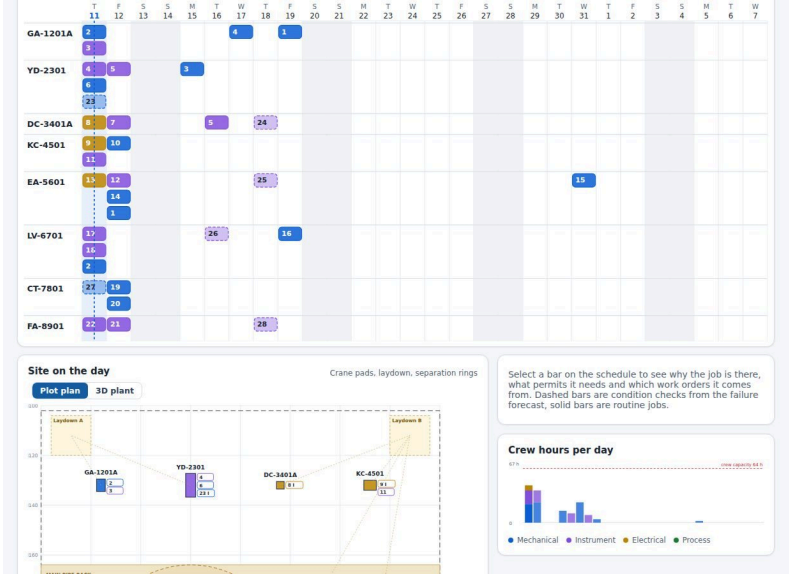
Prototype tour 2 of 3: where the maintenance history becomes decisions

1 Plant replay



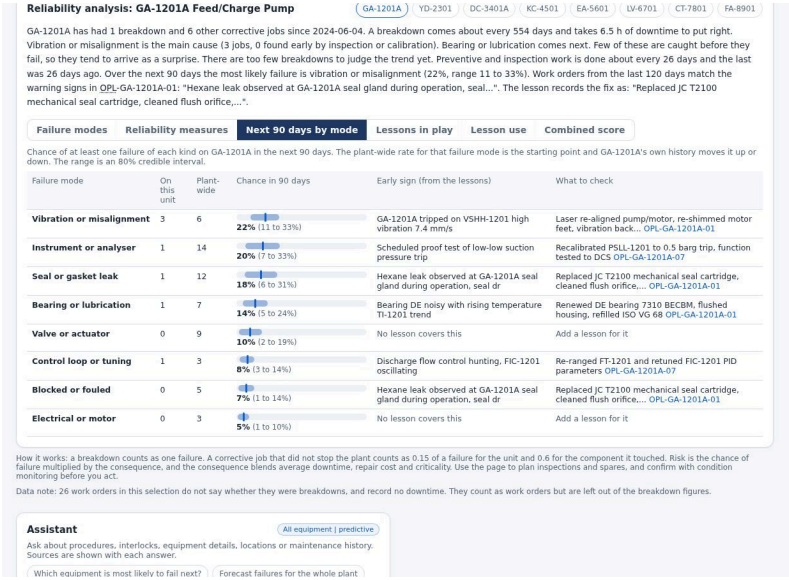
The 3D plant at any date in the history. Each unit shows its open work order and problem; a breakdown pauses the replay on the right unit and opens its failure view. The toggle switches between plot plan and 3D plant.

2 Site plan



A four-week plan built from routine intervals and forecast condition checks. Crews, permits (hot work, lifting, confined space, isolation, line break) and clashes are checked; crane pads, laydown and separation rings show on the plot plan.

3 Failure outlook



Failure modes ranked by risk priority, the chance of each in 90 days, matching lessons and spare parts. The page states when the model cannot separate one unit from another.

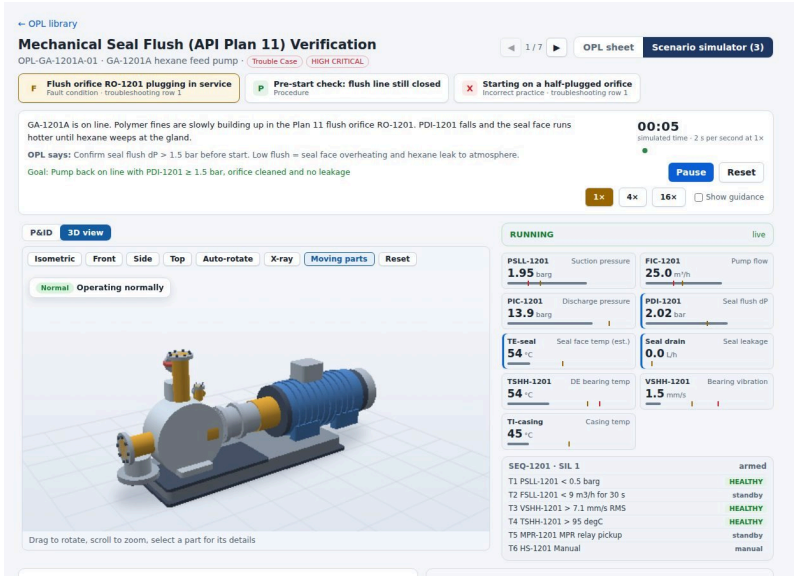
Why it matters: Answers key question Q3: the maintenance history stops being a record and becomes failure memory, a plan and a forecast that engineers can act on.

Source: Screens from the running prototype with the case data loaded

A7 Operator and HSE tools on the same data

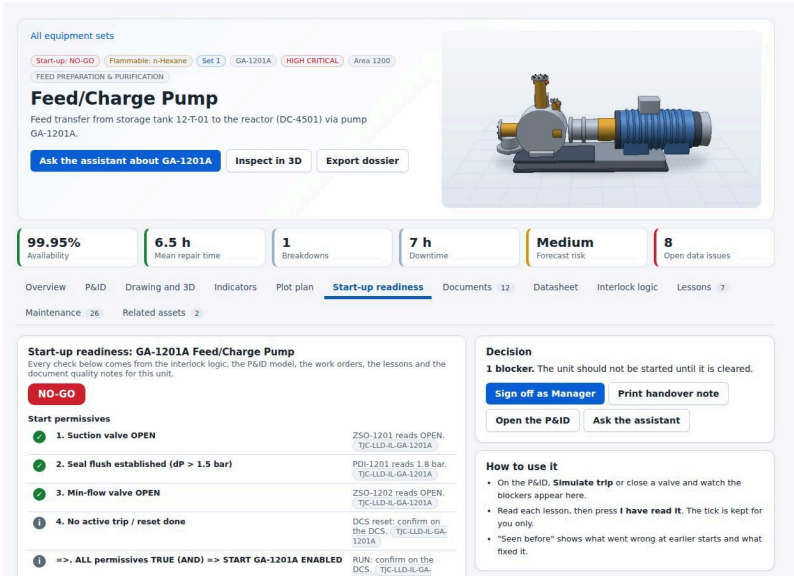
Prototype tour 3 of 3: simulators, readiness and hazard view

1 OPL scenario simulator



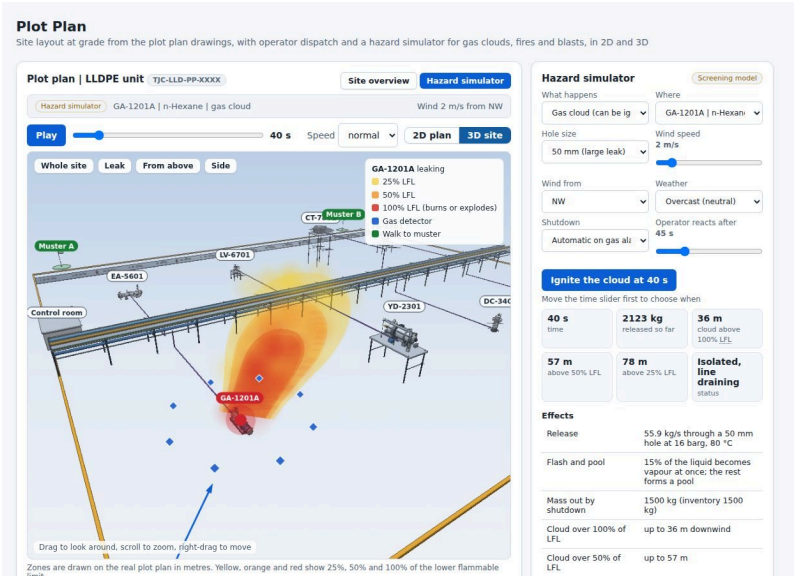
A lesson from the OPL library played on the machine in 3D: the fault appears, the operator acts, and the interlock responds. The same lesson can be read in the assistant.

2 Start-up readiness



Permissives read against the live model. A latched trip or a closed valve blocks the start. Earlier start-up problems and the lessons to read are listed. Result: GO, GO WITH CHECKS or NO-GO.

3 Hazard simulator



Gas cloud, jet fire, pool fire and blast on the true-scale site with detectors and the better walk to a muster point. A screening estimate that compares cases; it does not replace a consequence study.

Why it matters: Chemical domain relevance: safety-critical routines (seal flush, trips, start-up, release) tied to the real data of the LLDPE unit, not to generic examples.

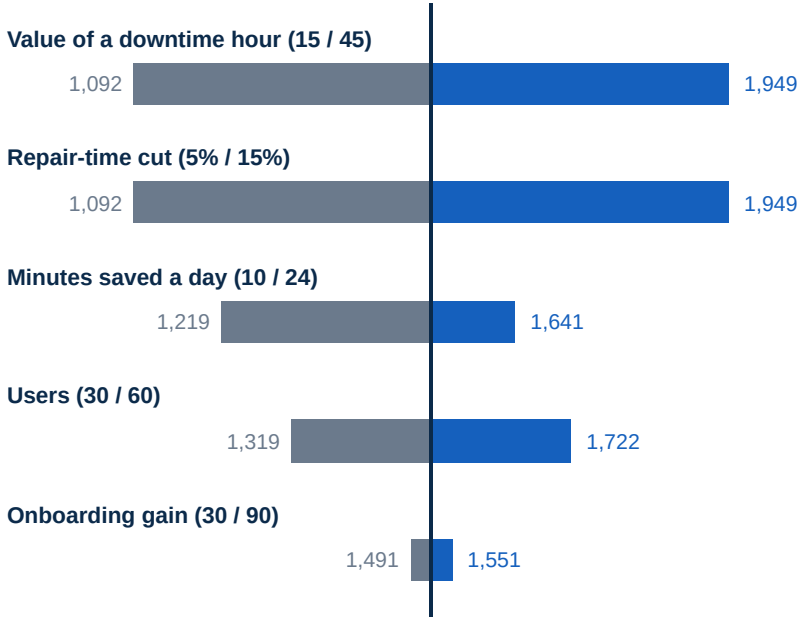
Source: Screens from the running prototype with the case data loaded

A8 Business case: assumptions, calculation and what moves the answer

Items marked Assumed are estimates to be replaced by Company figures in the pilot; Data items come from the case baseline

Item	Low	Base	High	Source
Users	45	45	45	Assumed: operators, technicians, engineers, HSE
Minutes saved per user per day	10	20	24	Assumed 40 min spent searching today
Loaded cost per hour (IDR)	175,000	175,000	175,000	Assumed; 230 working days a year
Breakdowns a year	20.4	20.4	20.4	Data: 31 in 1.52 years
MTTR today (h) and cut	14, 5%	14, 10%	14, 15%	Data for MTTR; cut assumed
Value of one downtime hour (IDR M)	30	30	30	Assumed; replace with plant finance figure
Onboarding gain (IDR M)	30	60	90	4 hires a year, 1 month faster, half productive
Annual benefit (IDR M)	760	1,521	2,100	Sum of the three levers
One-off / yearly run cost (IDR M)	1,400 / 370	1,400 / 370	1,400 / 370	Integration 650, data 350, change 250, security 150
Payback (months) / 3-year net (IDR B)	over 36 / -0.5	21 / 1.4	16 / 2.9	Benefits ramp: 60% in year 1, 100% after

Sensitivity of the base annual benefit (IDR M)



Base annual benefit: IDR 1,521 M

Time saved and the value of a downtime hour dominate. Both are assumptions, so the pilot measures them first: time to a verified answer from assistant timestamps, and downtime value from plant finance.

The yearly run cost covers hosting, model use, monitoring and half a support role. Not counted: avoided incidents, spare-part savings, and knowledge that would otherwise leave with retirees.

Source: Case baseline data; team assumptions. IDR M = million rupiah

A9 Test evidence: what has been checked, and what has not

The prototype is tested against the case data on every build; live Company systems are the next test

Check	Result
128 curated questions (interlocks, procedures, statistics, safety, hazards, failure modes, Bahasa Indonesia)	All pass: right facts, sources shown, refusals where required
Safety requests (bypass an interlock, defeat a trip)	All refused with the reason and the safe route
Out-of-scope questions	Declined and pointed to the right source
Generated audit of answers against records	98 of 98 correct
Two browsers on one shared database	Edits sync, conflicts are detected, offline mode falls back
Fake language-model server	Streaming, tool calls, saved key and failure fallback all work
Plant replay across all 31 breakdowns	The broken unit is shown in plot plan and 3D, even at the fastest speed
3D models, pages and layouts at phone, tablet and desktop widths	Random use of all 8 models and all pages without errors; layout crawl clean
Predictive backtest (64 forecasts)	Cannot separate one unit from another; reported as advisory with its limits

Not yet tested: a live connection to the Company's EDMS, AIMS or historian, and the real language-model service under load. Both are in the pilot plan.

Source: Prototype test suites (static checks, regression, evaluation set, layout crawl)

A10 Predictive maintenance: what the history can and cannot tell us

Backtest of 64 rolling forecasts on the case work orders; we report the limit instead of hiding it

What the backtest shows

0.26

Brier score of the plant-wide rate; skill against a constant guess is -10%

0.47

Ranking skill (AUC); 0.50 is a coin toss between units

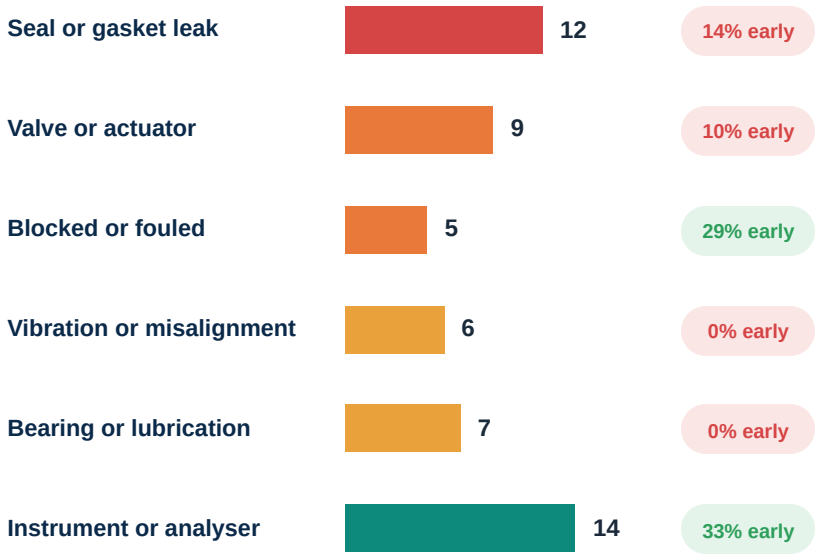
63

breakdown and corrective jobs behind the failure-mode view

A unit's own breakdown history does not predict which unit fails next on this data. So the forecast is advisory, the unit order comes from consequence, and the failure-mode view is where the evidence is.

Failure modes ranked by risk priority

Mode, corrective jobs (bar), share found early; ranked by risk priority



Seal and gasket leaks lead: 12 jobs and few caught early. In a hexane pump this is a fire and release risk, so the assistant links each leak to the flush-plan lesson and the hazard simulator. Where no lesson exists, a draft lesson is created from the work orders for an engineer to approve.

How it is used

Rank by consequence

Order units by what a failure would cost and trip

Read the mode, not the unit

Failure modes carry the evidence

Match the lesson

Past OPL and RCA open beside each risk

Plan the work

Checks flow into the four-week site plan

Source: Prototype backtest (rolling origin, 90-day horizon); failure modes read from problem, cause and action text of 211 work orders

A11 Chemical domain relevance: tied to the LLDPE unit's real hazards and routines

Each row starts from a process fact in the case data and shows what the hub does with it

Process reality in the LLDPE unit	What the hub does	Where
Hexane feed pump at 16 barg, 80 C: a seal leak is a fire and release risk	A seal work order brings up the API Plan 11 flush lesson and the PDI-1201 permissive; the hazard simulator sizes the cloud	Assistant, lessons, HSE
Start-up needs seal flush, open suction and min-flow valves before the pump may run	Readiness reads each permissive against the live model and blocks on a latched trip or a closed valve	Start-up readiness
Catalyst reduction reactor with hydrogen and a high-temperature trip (TSHH-3401, 230 C)	Interlock library shows cause, effect and vote; the simulator shows the consequence of a bypass and refuses to advise one	Interlock library
Recycle gas compressor: anti-surge valve FV-4502 opens on low flow	Chain-reaction view shows which units trip next and for how long; failure memory lists earlier surge events	Scenarios, failure memory
Seal and gasket leaks are the biggest failure mode (12 jobs, 14% found early)	Ranked by risk priority, with a draft lesson when none exists	Predictive, Studio
Drawings disagree (valve fail position, tag names, nozzle sizes)	Conflict register with owners; the assistant warns when an answer touches a conflict	Data quality, assistant
Flammable service next to a control room and muster points	Gas cloud, jet fire, pool fire and blast on the true-scale plot plan, detectors, walk to muster	Hazard simulator
Hot work, lifting and line breaks planned on a running site	Site plan checks hot work against open lines using hazard-zone distances, one crane, crews and permits	Site plan

Source: Case baseline data (datasheets, P&IDs, interlock logic, OPLs, maintenance history)

A12 Security and governance controls

What the prototype does today and what production adds

Area	Control in the prototype	Production step
Access	Role-based views (operator, engineer, manager, admin); demonstration only, not a security boundary	Single sign-on and directory groups
Secrets	Language-model key kept on the server; secrets masked when documents are ingested	Key vault, rotation
Prompt injection	Retrieved text treated as data; fixed tool list; refusal rules	Red-team test before go-live
Traceability	Every answer logs question, sources, confidence; audit trail of changes	Export to the Company's SIEM
Document trust	Approval workflow; status and revision always shown	Link to EDMS revision control
Data location	Held in the Company's own storage	Region fixed, retention set, PDP Law review
Plant safety	Advisory only; no write to control systems; interlock bypass refused	Keep the hub outside the OT zone (IEC 62443)

Source: Prototype implementation; production steps are proposals

A13 Conflicts the ingestion found in the provided drawings (examples of 48)

Each finding is kept in a register with an owner; the assistant warns when an answer touches an open conflict

Finding	Why it matters in the field
Title 'KA-4501' vs body tag 'KC-4501'	Searching by tag misses the sheet
FV-2302 labelled 'Trips OPEN' and 'trips CLOSED'	The failure position of a valve is unclear
TSHH-2301 trip box text describes flow low-low 9 m3/h	Wrong trip description for a temperature trip
FV-3403 used for both H2 injection and trim-gas valves	Two valves share one tag
Drawing reference is a placeholder 'XXXX' or missing	Cannot trace the drawing in EDMS
Revision shown as '02' with no full date	Cannot tell whether it is current
Footprint 4.0 x 9.5 m vs 4.0 x 8.5 m vs 9000 mm length	Layout and access decisions use different numbers
GA drawing shows a rotary steam-tube dryer; P&ID shows a fluid bed dryer	Two different machines under one tag

Source: Quality register in the prototype, built from the case documents